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REVIEW

Efficacy and Safety of Nerve Ablation Techniques Versus Sham Procedures in Knee Osteoarthritis: A Systematic Review and Meta-Analysis of Randomized Clinical Trials

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Received: 22 March 2025 | **Accepted:** 11 August 2025

Funding: The authors received no specific funding for this work.

Keywords: arthralgia | arthritis | joint pain | neurolysis | pain | pulsed radiofrequency | radiofrequency ablation

ABSTRACT

Background: Ablative techniques, such as radiofrequency (RF) and cryoneurolysis (CN), are emerging as minimally invasive options for knee osteoarthritis (OA) pain management. However, inconsistencies in efficacy, technique variations, and a lack of standardized protocols limit their clinical application. This study addresses these gaps through a comprehensive meta-analysis.

Objectives: This systematic review and meta-analysis aimed to evaluate the efficacy of ablative techniques in improving (1) pain, measured by the Visual Analog Scale (VAS), (2) functionality, assessed using the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and (3) quality of life and patient perception/satisfaction at 4, 12, and 24 weeks after treatment in patients with OA.

Methods: A meta-analysis of randomized controlled trials (RCTs) was conducted using PubMed, Embase, Scopus, and the Cochrane Database up to October 2024. Studies were included evaluating genicular or saphenous nerve ablation in adults with OA-associated pain. Outcomes of interest comprised pain reduction (VAS), functional improvement (WOMAC), patient perception, and quality of life. The risk of bias was evaluated using the RoB 2 tool, and data were analyzed using fixed- or random-effects models based on the degree of heterogeneity.

Results: Seven studies involving a total of 437 patients were included. The meta-analysis demonstrated a significant reduction in pain in the ablation group at 4 weeks (MD = −1.60; 95% CI = −2.79 to −0.41), 12 weeks (MD = −1.97; 95% CI = −2.54 to −0.80), and 24 weeks (MD = −0.82; 95% CI = −1.41 to −0.23). CN consistently reduced pain (MD = −2.25 at 4 weeks, −1.12 at 12 weeks, and −1.28 at 24 weeks). Additionally, most evaluations did not show significant differences in patient perception and quality of life measures.

Conclusion: Ablative techniques, particularly CN, show promise in reducing pain and enhancing function in patients with OA, offering a minimally invasive alternative for those with an inadequate response to conservative treatments. However, the observed outcome variability underscores the need for further research to standardize protocols and confirm long-term safety and efficacy.

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1 | Introduction

Knee osteoarthritis (OA) is a progressive degenerative disease affecting at least 19% of American adults aged 45 and older, representing one of the leading causes of disability and pain worldwide [1]. Among its various symptoms, such as stiffness and insomnia, the chronic pain associated with knee OA significantly impacts quality of life (QoL) [2, 3].

Several non-surgical treatment modalities have been explored with the aim of postponing or even potentially avoiding the need for surgical intervention. Among these modalities, platelet-rich plasma administration, intra-articular hyaluronic acid injections, shockwave therapy, orthotic support through braces, and, more recently, the application of ablative techniques (ATs) are notable [4–7]. However, these techniques still lack robust evidence and require further evaluation regarding their effectiveness.

Introduced in 2010, the ablative radiofrequency (RF) approach has emerged as a minimally invasive option for managing knee OA [7]. This technique has the potential to enhance functionality and alleviate pain in patients who do not benefit from conservative management strategies or are not candidates for total knee arthroplasty [8]. Although clinical studies have shown promising results, the variety of techniques and the novelty of the procedure raise uncertainties about its application [8, 9]. Currently, there are no guidelines or consensus recommending the use of ATs in the management of knee OA [10].

This meta-analysis aims to evaluate the efficacy of ATs in reducing pain associated with knee OA, comparing them with sham procedures. By providing a rigorous analysis, this study offers a substantial reference point for clinical practice and guides future research in this field.

2 | Methods

This systematic review and meta-analysis was conducted and written by the Cochrane Collaboration Handbook for Systematic Reviews of Interventions and the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [11]. The aim was to evaluate the efficacy of genicular or saphenous nerve ablation in reducing pain in patients with knee osteoarthritis. The review followed the PICO framework (Population, Intervention, Control, Outcome) with the following inclusion criteria: *P* = Patients with pain associated with knee osteoarthritis; I = Genicular or saphenous nerve ablation; C = Sham treatment; O = Reduction in pain intensity, improvement in functionality, quality of life, patient perception, and adverse events. This study was registered in the International Prospective Register of Systematic Reviews (PROSPERO) under registration number CRD42024498470. The methodology was defined and registered before the database search began.

2.1 | Eligibility Criteria

We included full-text randomized clinical trials published in English from inception to October 2024, available in the databases PubMed, Embase, Scopus, and Cochrane Central Register of Controlled Trials. These studies had to include patients aged 18 years or older with chronic osteoarthritis, compare ablation techniques (ATs) with sham procedures, and report at least one of the clinical outcomes of interest.

Exclusion criteria comprised articles lacking sham procedures, conference abstracts, case reports or case series, preprints, observational studies, letters to the editor, nonrandomized trials, literature reviews, studies not published in English, and studies with insufficient data on relevant outcomes. Duplicate publications of the same study were also excluded; only the most comprehensive report was included.

2.2 | Search Strategy

For the database search strategy, descriptors related to "ablation," "genicular nerve," "saphenous nerve," "radiofrequency," "cooling," and "osteoarthritis" were used, available in their respective descriptor repositories. The descriptors used in PubMed, Scopus, EMBASE, and Cochrane were obtained from the Medical Subject Headings (MeSH), accessed via www.ncbi.nlm.nih.gov/mesh/. Boolean descriptors '*AN*' and 'OR' were used for term searches on the mentioned platforms, adhering to the articles' inclusion and exclusion criteria. All descriptors used, and the complete search strategy for each database are available in Data S1.

2.3 | Selection of Studies

Two reviewers (RCOP and CRAJ), who had no access to each other's assessments, evaluated the identified article titles, selecting those that met the inclusion criteria. Subsequently, abstracts were reviewed for selection purposes. Articles that passed the initial screening were examined to determine inclusion in the systematic review. When discrepancies arose between the two reviewers, a senior reviewer (DCK), who had exclusive access to the conflicting articles, intervened to resolve differences. The selection of the studies was made using the Rayyan app [12].

2.4 | Data Summarization

Data extraction was performed independently and in duplicate (RCOP and CRAJ) to ensure accuracy and reliability. Two reviewers extracted data from the included studies using a pre-designed data extraction form created in Microsoft Excel (version 2205). The form captured comprehensive details on study characteristics, including population size, intervention and control group specifics, methodological approaches, and the outcomes assessed. Any discrepancies were resolved through discussion or consultation with a senior reviewer (DCK) to achieve consensus.

2.5 | Quality Assessment

The risk of bias assessment was conducted using the Cochrane Risk of Bias tool for randomized trials (RoB 2) [13], evaluated

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independently by two reviewers (ESRB and JPFG), with discrepancies resolved through a senior reviewer (DCK). This tool addresses five specific domains for bias assessment, including randomization, potential deviations in the intervention, absence of data, outcome measurement, and result selection. After each reviewer evaluates each domain, risks are categorized as 'low risk' 'high risk' or some concerns indicating an uncertain bias risk, and the overall bias risk of the article is calculated. The funnel-plot analysis of point estimates about study weights 14 investigated publication bias.

2.6 | Statistical Analysis

Forest plots for continuous outcomes were constructed based on mean and standard deviation (SD). When results were reported as median and interquartile range, the values were converted to mean and SD using the methods described by Luo et al. [14] and Wan et al. [15], respectively.

Pain reduction, a continuous outcome measured using the Visual Analog Scale (VAS), and functional assessment, calculated using the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), were analyzed using the mean difference (MD) and presented with 95% confidence intervals (CIs). Patient perception and QoL, assessed using different scales for these outcomes, were analyzed using the standardized mean difference (SMD), also with 95% CIs. The analysis of outcomes was stratified based on the technique used for target nerve ablation.

In studies where an increase in the mean and standard deviation indicated an improvement in the outcome, a negative sign was used to ensure increased values reflected a better response to treatment. This approach ensures consistency in interpreting results, especially when the scales used may have opposite directions regarding the desired effect.

Heterogeneity among the studies was assessed using the Cochran Q test and the I^2 statistic, with $p < 0.10$ and $I^2 > 25\%$ indicative of significant heterogeneity. A fixed-effects model was applied for outcomes with low heterogeneity ($I^2 \leq 25\%$); in contrast, for outcomes with higher heterogeneity, the random-effects model of DerSimonian and Laird was used.

A leave-one-out sensitivity analysis evaluated each study's influence on the overall forest plot. All statistical analyses were conducted using Review Manager 5.4 (Cochrane Center, The Cochrane Collaboration).

3 | Results

A database search yielded 224 articles. After the removal of duplicates, 132 articles remained. A screening of titles and abstracts resulted in selecting 21 articles for full-text review. Finally, seven articles were included in the review [7, 16–21]. All studies were included in the meta-analysis. The study selection process and the reasons for exclusion are shown in Figure 1.

3.1 | Study Characteristics

This review included seven studies, all of which were randomized, double-blind, sham-controlled clinical trials. A total of 437 patients were evaluated, with 249 receiving ablation treatment. In two studies, the pulsed RF technique was employed, with all applying a temperature of up to 42°C and a duration of 6 min [18, 19]. Three other studies utilized conventional RF, with temperatures ranging from 70°C to 90°C and 90 to 180-s durations [7, 16, 17]. Only one study evaluated cooled RF, using a temperature of 60°C for 90 s [21]. Finally, one study applied cryoneurolysis (CN) with nitrous oxide for 23 min at −20°C [20]. The follow-up period ranged from 12 to 52 weeks. All studies included in this review assessed pain reduction as the primary outcome using the VAS. At baseline, pain levels in the intervention group ranged from 6.62 ± 1.57 to 8.25 ± 0.70, while in the control group, values ranged from 6.3 ± 1.56 to 7.72 ± 0.75. Further details about the study protocols and their main characteristics are available in Tables 1 and 2.

3.2 | Pain Reduction Using the VAS

3.2.1 | 4-Week Follow-Up

The meta-analysis included six studies totaling 416 participants for this outcome [7, 17–21]. The overall analysis indicated a significant reduction in pain favoring the ablation group (MD = −1.60; 95% CI = −2.79 to −0.41; $p = 0.002$; $I^2 = 84\%$) (Figure 2). Analysis by technique revealed that pulsed approaches (MD = −1.55; 95% CI = −3.09 to 0.00; $p = 0.05$; $I^2 = 60\%$) and CN (MD = −2.25; 95% CI = −3.00 to −1.30; $p < 0.0001$) were effective in reducing pain, favoring the ablation group. In contrast, conventional (MD = −1.97; 95% CI = −5.64 to 1.71; $p = 0.29$; $I^2 = 96\%$) and cooled techniques (MD = −0.20; 95% CI = −1.66 to 1.26; $p = 0.79$) did not show a significant reduction. Funnel plot asymmetry was observed (Figure S1, Data S2). The sensitivity analysis indicated that the exclusion of Radnovich et al. [20] resulted in the loss of significance in pain reduction (MD = −1.47; 95% CI = −3.00 to 0.06; $p = 0.06$; $I^2 = 87\%$) (Figure S2, Data S2).

3.3 | 12-Week Follow-Up

Five studies, including 352 participants, were analyzed for this outcome [7, 18–21]. The overall analysis demonstrated a significant reduction in pain in the ablation group (MD = −1.97; 95% CI = −2.54 to −0.80; $p = 0.002$; $I^2 = 64\%$) (Figure 3). Technique-specific evaluation showed significant reductions for conventional (MD = −3.53; 95% CI = −4.79 to −2.27; $p < 0.00001$), pulsed (MD = −1.21; 95% CI = −2.22 to −0.21; $p = 0.02$; $I^2 = 0\%$), cooled (MD = −1.30; 95% CI = −2.21 to −0.39; $p = 0.005$), and CN approaches (MD = −1.12; 95% CI = −2.07 to −0.17; $p = 0.02$), all favoring the ablation group. Funnel plot asymmetry was detected (Figure S3, Data S2). Sensitivity analyses did not reveal significant changes in the overall results.

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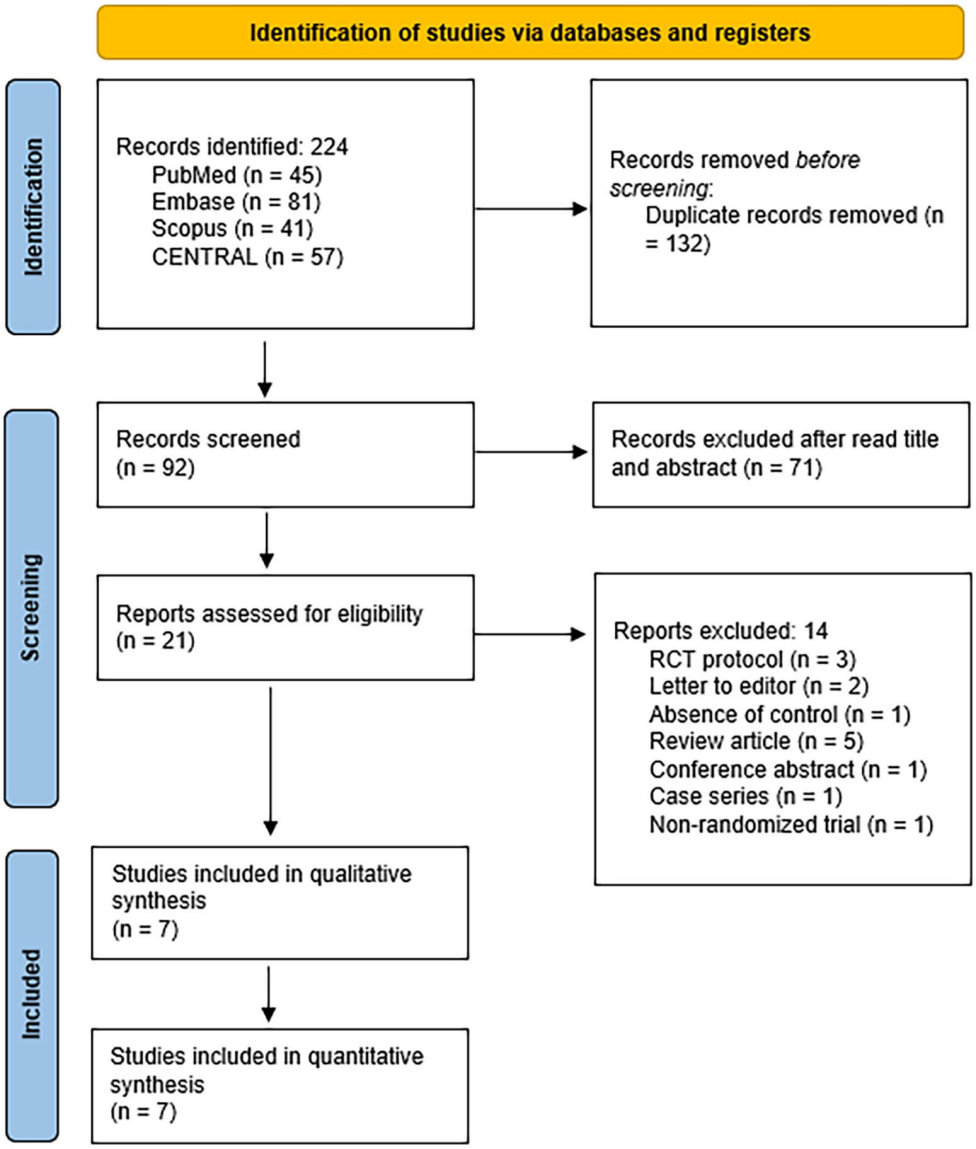

FIGURE 1 | Flowchart of included studies.

3.4 | 24-Week Follow-Up

Four studies, totaling 306 participants, were included in the meta-analysis [7, 17, 18, 20]. The overall analysis indicated a significant reduction in pain in the ablation group compared to sham (MD = −0.82; 95% CI = −1.41 to −0.23; $p = 0.002$; $I^2 = 20\%$) (Figure 4). In the technique-specific analysis, only CN significantly reduced pain (MD = −1.28; 95% CI = −2.30 to −0.26; $p = 0.01$). Conventional (MD = 0.20; 95% CI = −1.08 to 1.48; $p = 0.76$), pulsed (MD = −0.70; 95% CI = −1.88 to 0.48; $p = 0.24$), and cooled techniques (MD = −1.30; 95% CI = −2.64 to 0.04; $p = 0.06$) did not show significant reductions. Funnel plot asymmetry was observed for this outcome (Figure S4, Data S2). The sensitivity analysis indicated that the exclusion of Radnovich et al. [20] resulted in a loss of significance in pain reduction (MD = −0.59; 95% CI = −1.31 to 0.14; $p = 0.28$; $I^2 = 22\%$) (Figure S5, Data S2).

3.5 | Change in Total WOMAC Score

3.5.1 | 4-Week Follow-Up

Four studies, totaling 307 participants, were included in this outcome [16, 17, 20]. The overall analysis indicated a significant reduction in the total WOMAC score between the ablation and sham groups (MD = −5.82; 95% CI = −9.37 to −2.28; $p = 0.001$; $I^2 = 22\%$) (Figure S6, Data S2). Technique-specific analysis revealed that only the CN approach (MD = −7.10; 95% CI = −11.04 to −3.16; $p = 0.0004$) showed a significant reduction, favoring the ablation group. In contrast, conventional (MD = −3.20; 95% CI = −26.39 to 19.99; $p = 0.79$), pulsed (MD = −4.04; 95% CI = −14.83 to 6.69; $p = 0.46$), and cooled techniques (MD = 7.80; 95% CI = −6.97 to 22.57; $p = 0.30$) did not show significant reductions. The sensitivity analysis indicated that the exclusion of Radnovich et al. [20] resulted in the loss of significance in

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TABLE 1 | The main characteristics of the included studies.

Authors/ Year	Intervention group (*n*)	Control group (*n*)	Gender (*n*)[a]	K-L grade	Evaluation of pain	Baseline pain[b]	BMI (kg/m^2)[b]
Choi et al. [7]	17	18	5/30	II-IV	VAS	Intervention: 7.82 ± 1.38 Control: 7.72 ± 0.75	Intervention: 26.2 ± 3.3 Control: 26.5 ± 2.1
Radnovich et al. [20]	121	59	61/119	II-III	VAS	Intervention: 6.62 ± 1.57 Control: 7.09 ± 1.56	Intervention: 29.1 ± 3.97 Control: 30.0 ± 3.54
Tabasco et al. [16]	12	16	7/18	III-IV	VAS	Intervention: NA Control: NA	Overall: 29.9 ± 4.64
Uematsu et al. [19]	37	33	14/56	II-IV	VAS	Intervention: 7.0 ± 1.92 Control: 7.5 ± 1.59	Intervention: 25.99 ± 3.38 Control: 24.44 ± 3.72
Carpenedo et al. [18]	10	10	7/13	III-IV	VAS	Intervention: 8.25 ± 0.70 Control: 8.0 ± 1.19	Intervention: 29.48 ± 4.42 Control: 29.62 ± 3.43
Malaithong et al. [17]	32	32	12/52	III-IV	VAS	Intervention: 7.1 ± 1.7 Control: 6.3 ± 1.7	Intervention: 29 ± 3 Control: 28 ± 5.0
Kwon et al. [21]	20	20	10/30	II-IV	VAS	Intervention: 6.8 ± 1.6 Control: 6.3 ± 1.6	Intervention: 25.4 ± 3.3 Control: 26.7 ± 4.0

Abbreviations: BMI, body mass index; K-L, Kellgren-Lawrence; VAS, Visual Analogue Scale.
[a]Male/female.
[b]Mean ± standard deviation.

the WOMAC reduction (MD = −0.36; 95% CI = −8.50 to 7.79; $p = 0.93$; $I^2 = 0\%$) (Figure S7, Data S2).

3.5.2 | 12-Week Follow-Up

Three studies, totaling 243 participants, were included in the meta-analysis for this outcome [16, 20, 21]. The overall analysis showed a significant reduction in WOMAC scores between the ablation and sham groups (MD = −3.60; 95% CI = −6.92 to −0.28; $p = 0.03$; $I^2 = 0\%$) (Figure S8, Data S2). Technique-specific analysis revealed that only CN (MD = −4.72; 95% CI = −8.55 to −0.89; $p = 0.02$) showed a significant difference, favoring the ablation group. Pulsed (MD = −1.20; 95% CI = −9.01 to 6.61; $p = 0.30$) and cooled techniques (MD = 2.20; 95% CI = −10.36 to 14.76; $p = 0.73$) did not show significant differences. The sensitivity analysis showed that the exclusion of Radnovich et al. [20] resulted in the loss of statistical significance in the overall analysis (MD = −0.25; 95% CI = −6.88 to 6.38; $p = 0.94$; $I^2 = 0\%$) (Figure S9, Data S2).

3.5.3 | 24-Week Follow-Up

Three studies, with 255 participants, were included in the meta-analysis for this outcome [17, 20, 21]. The overall analysis indicated a significant reduction in WOMAC scores between the ablation and sham groups, favoring the ablation group (MD = −3.26; 95% CI = −5.10 to −1.41; $p = 0.0005$; $I^2 = 8\%$) (Figure S10, Data S2). Only CN (MD = −3.26; 95% CI = −5.10 to −1.41; $p = 0.0005$) showed a significant difference between the groups. Conventional (MD = 3.30; 95% CI = −26.12 to 32.72; $p = 0.83$) and cooled techniques (MD = 7.20; 95% CI = −7.54 to 21.94; $p = 0.34$) did not show significant differences. The sensitivity analysis revealed that the exclusion of Radnovich et al. [20] resulted in the loss of significance in the overall analysis (MD = 6.42; 95% CI = −6.76 to 19.59; $p = 0.34$; $I^2 = 0\%$) (Figure S11, Data S2).

3.6 | Patient Perception

To assess patients' perception of change regarding treatment, the study by Choi et al. [7] employed the Global Perception of Change (GPE) scale, while Malaithong et al. [17] used the Patient Global Impression of Improvement (PGI-I), and Kwon et al. [21] utilized the Global Perceived Effect of Satisfaction (GPES). These instruments generally evaluate the patient's subjective progress or satisfaction with treatment outcomes, including pain relief, functional improvement, or overall well-being. All these scales use a 1-to-7-point scoring system, where higher values indicate greater patient satisfaction. However, the study by Radnovich et al. [20] employed the Patient Global Impression of Change (PGI-C) scale, which also ranges from 1 to 7 points but has an inverted interpretation. In this case, higher scores indicate lower patient satisfaction.

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TABLE 2 | Protocol of the included studies.

Authors/Year	Population	Age (Years)*	Control	Intervention	Follow-up time
Choi et al. [7]	Patients with chronic knee pain for at least 3 months, diagnosed with OA, who have not responded to alternative treatments.	Intervention: 67.9 ± 7.1 Control: 66.5 ± 4.8	Sham	Percutaneous genicular nerve neurotomy using RF at 70°C for 90 s. Modality: Conventional Polar: Monopolar Imaging: FS	12 weeks
Radnovich et al. [20]	Patients who met the clinical and radiographic criteria established by the ACR for knee OA.	Intervention: 60.6 ± 8.98 Control: 61.3 ± 8.17	Sham	CN of the infrapatellar saphenous nerve through percutaneous cryotherapy from −20°C to −100°C for 23 min. Modality: Nitrous oxide Polar: NA Imaging: NA	26 weeks
Tabasco et al. [16]	Patients diagnosed with knee OA and experiencing chronic pain for more than 3 months, with no response to conservative treatments.	Intervention: 75.2 ± 9.1 Each group's means and standard deviations were not specified; only the values for all participants were provided.	Sham	RF with temperature control (3 cycles of 2 min) on the saphenous nerve, followed by RF ablation of the genicular nerve at 80°C for 90 s. Modality: Conventional Polar: Monopolar Imaging: US	26 weeks
Uematsu et al. [19]	Patients diagnosed with OA through radiological examinations, experiencing persistent pain for more than 3 months, and unresponsive to conservative treatments.	Intervention: 73.05 ± 10.12 Control: 74.30 ± 8.52	Sham	RF ablation of the saphenous nerve in 4 cycles of 120 s each, with temperature below 42°C. Modality: Pulsed Polar: NA Imaging: US	12 weeks
Carpenedo et al. [18]	Patients diagnosed with chronic knee OA, experiencing pain for at least 6 months, and unresponsive to conservative treatments.	Intervention: 70.37 ± 7.36 Control: 70.87 ± 11.81	Sham	RF ablation of the saphenous nerve in 4 cycles of 120 s each, with temperature below 42°C. Modality: Pulsed Polar: NA Imaging: US	24 weeks
Malaithong et al. [17]	Patients with severe OA, chronic knee pain for more than 3 months, and failure of conservative treatments.	Intervention: 64.8 ± 11.1 Control: 68.6 ± 8.5	Sham	RF ablation of the genicular nerve at 90°C for 180 s. Modality: Conventional Polar: Bipolar Imaging: FS	52 weeks
Kwon et al. [21]	Patient with chronic unilateral knee pain due to OA for over 6 months, confirmed by radiological examinations and refractory to conservative treatment.	Intervention: 69.1 ± 11.8 Control: 71.9 ± 6.0	Sham	RF ablation of the genicular nerve at 60°C for 90 s. Modality: Cooled Polar: NA Imaging: FS	26 weeks

Abbreviations: ACR, American College of Rheumatology; CN, cryoneurolysis; FS, fluoroscopy; NA, not available; OA, osteoarthritis; RF, radiofrequency; US, ultrasound.
*Mean ± standard deviation.

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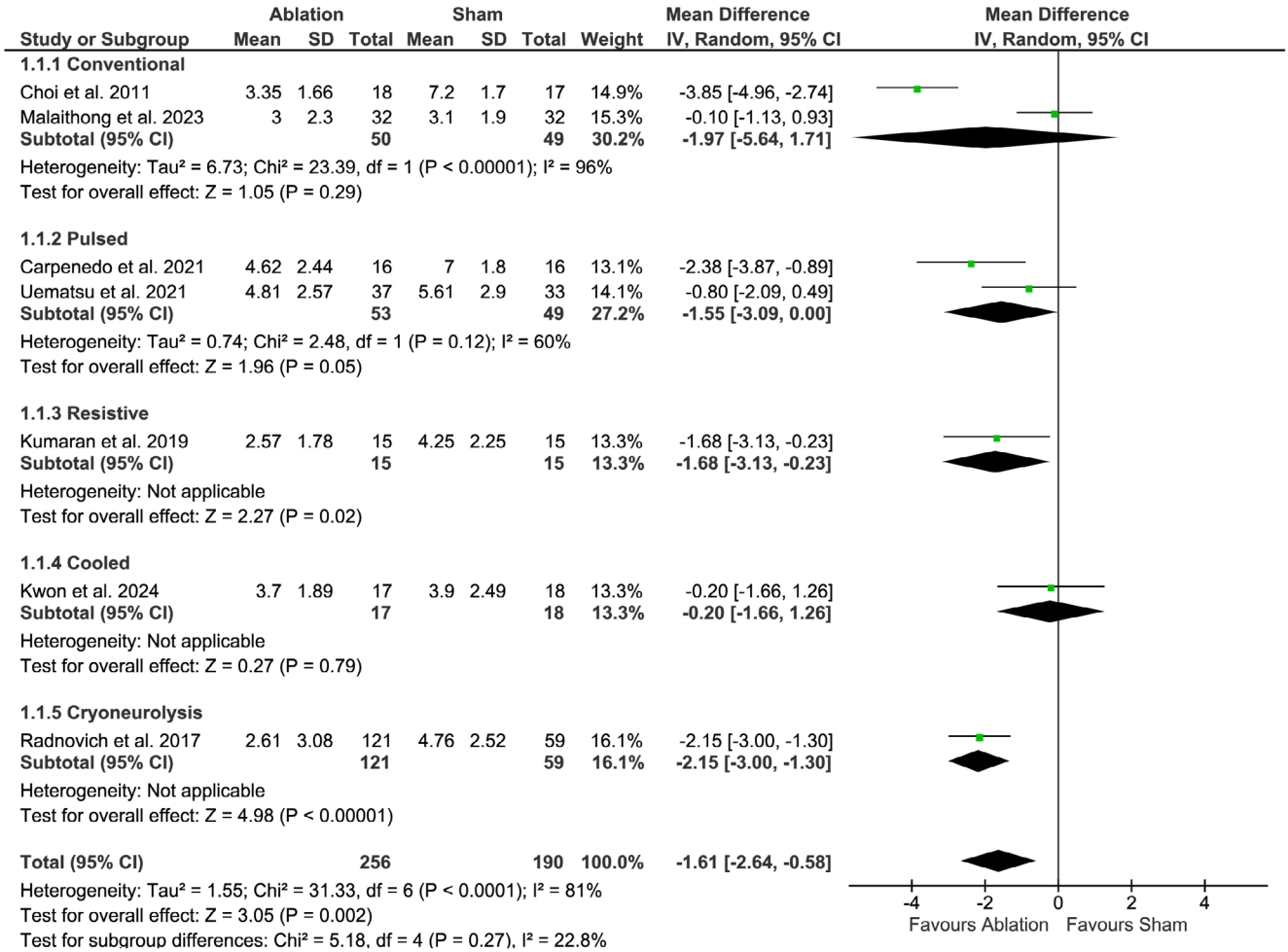

FIGURE 2 | Forest plot evaluating the pain in the 4-week follow-up.

3.6.1 | 4-Week Follow-Up

Four studies, totaling 310 participants, were included in this meta-analysis for this outcome [7, 17, 20, 21]. The overall analysis found no statistically significant difference between the ablation and sham groups (SMD = 0.66; 95% CI = −0.22 to 1.55; $p = 0.14$; $I^2 = 91\%$) (Figure S12, Data S2). Neither the conventional ATs (SMD = 0.77; 95% CI = −1.27 to 2.81; $p = 0.46$; $I^2 = 95\%$) nor the cooled ATs (SMD = 0.00; 95% CI = −0.66 to 0.66; $p = 1.00$) showed significant differences between the groups. However, CN demonstrated a significant effect, favoring ablation over sham (SMD = 1.14; 95% CI = 0.80 to 1.48; $p < 0.00001$). In the sensitivity analysis, the exclusion of Malaithong et al. [17] resulted in a significant difference between the groups, favoring the ablation group (MD = 0.98; 95% CI = 0.09 to 1.86; $p = 0.03$; $I^2 = 85\%$) (Figure S13, Data S2).

3.6.2 | 12-Week Follow-Up

Four studies, totaling 233 participants, were included in this meta-analysis for this outcome [7, 17, 20, 21]. The overall analysis did not demonstrate a statistically significant difference between the ablation and sham groups (SMD = 0.74; 95% CI = −0.15 to 1.63; $p = 0.10$; $I^2 = 90\%$) (Figure S14, Data S2). Both the conventional ATs (SMD = 0.92; 95% CI = −1.31 to 3.15; $p = 0.42$; $I^2 = 95\%$) and CN (SMD = 0.03; 95% CI = −0.29 to 0.35; $p = 0.87$) showed no significant differences between the groups. On the other hand, the cooled ablation technique demonstrated a significant effect, favoring the ablation group (SMD = 1.31; 95% CI = 0.57 to 2.05; $p = 0.0005$). Sensitivity analyses did not indicate any relevant variations in the overall result.

3.7 | Quality of Life

To assess the QoL of participants, the studies by Radnovich et al. [20] and Carpenedo et al. [18] used the Short Form Health Survey (SF-36). This questionnaire measures eight physical and mental health dimensions, with higher scores indicating better QoL. Meanwhile, the study by Uematsu et al. [19] employed the QoL subscale of the Knee Injury and Osteoarthritis Outcome Score (KOOS), which evaluates the impact of knee injuries and osteoarthritis, with higher scores also indicating better QoL.

3.7.1 | 4-Week Follow-Up

Three studies, totaling 270 participants, were included in this meta-analysis for this outcome [18–20]. The overall analysis did not show a statistically significant difference between the ablation and sham groups (SMD = 0.07; 95% CI = −0.18 to 0.32;

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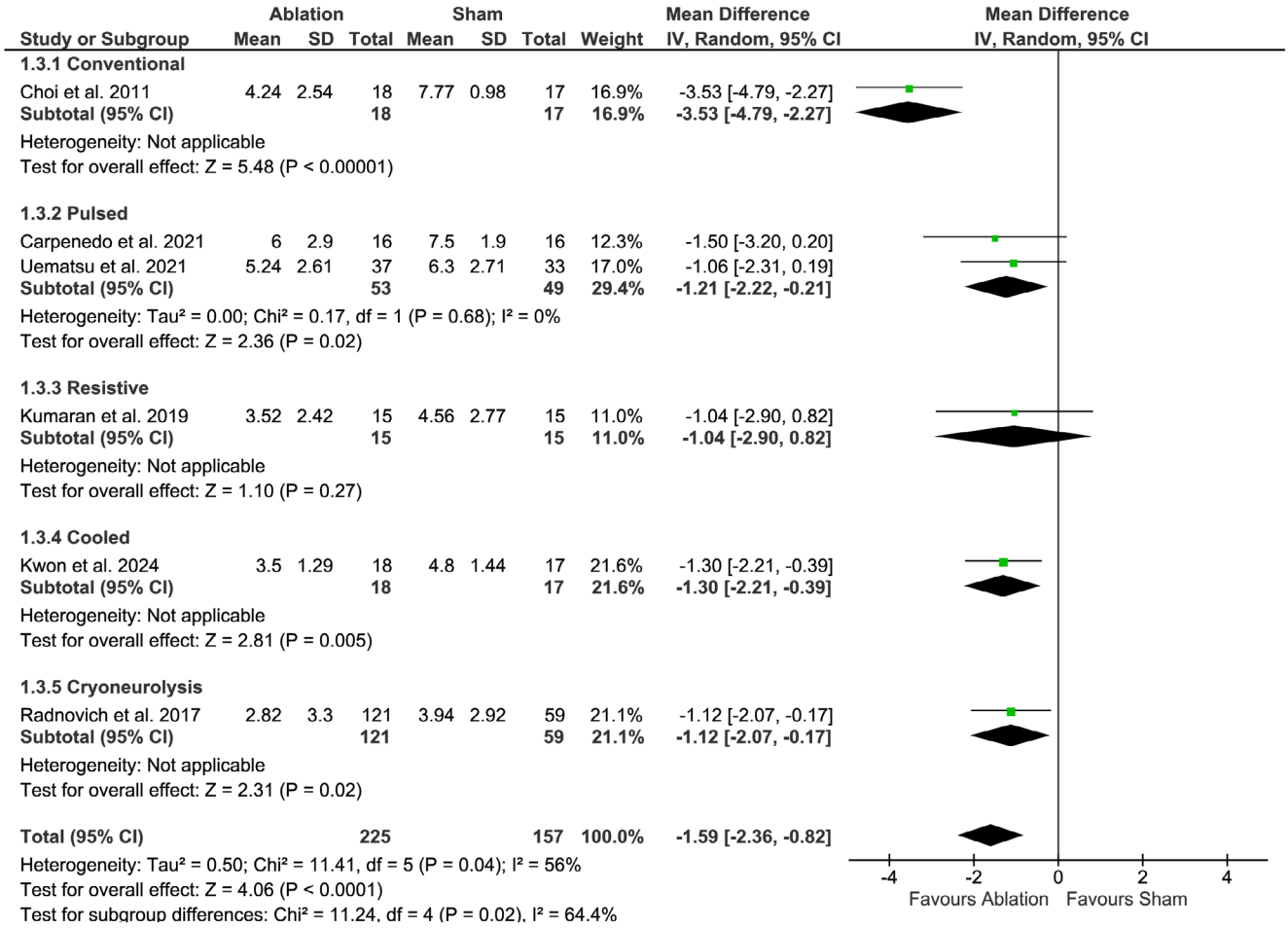

FIGURE 3 | Forest plot evaluating the pain in the 12-week follow-up.

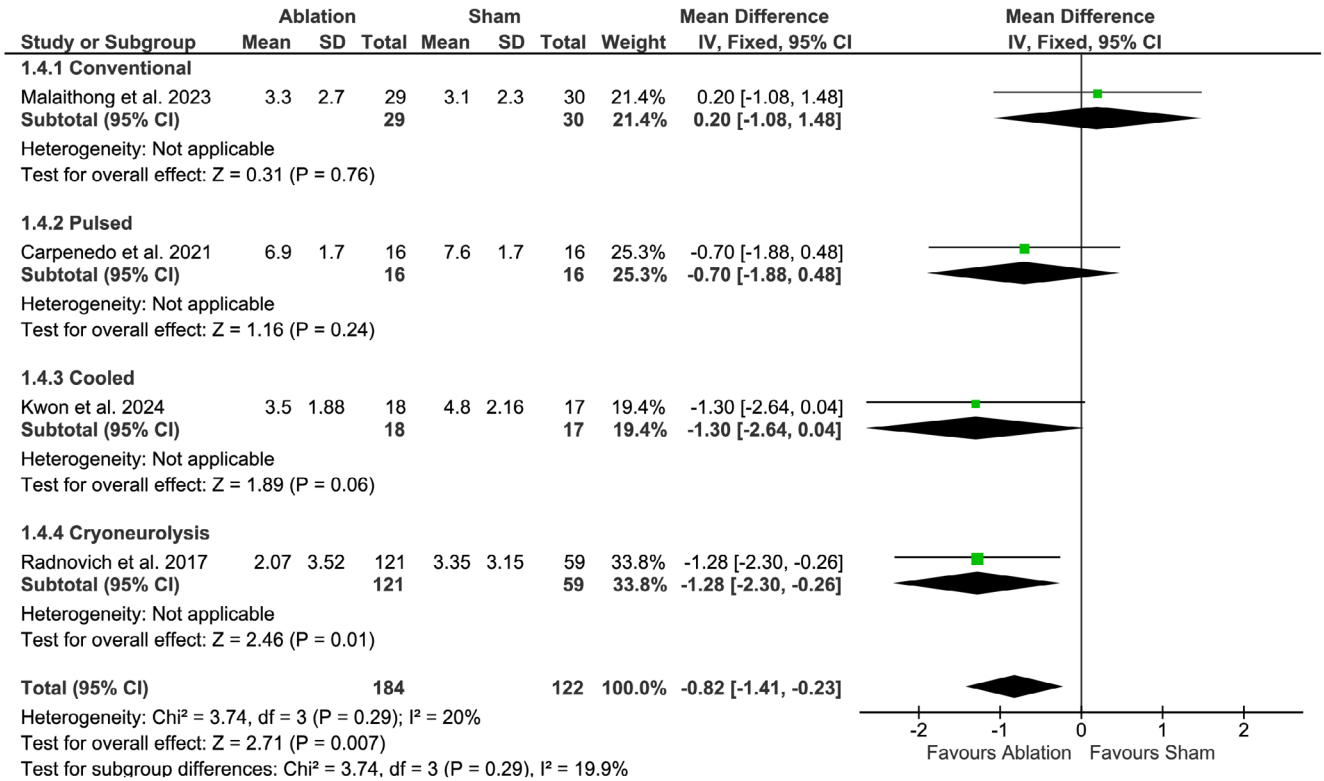

FIGURE 4 | Forest plot evaluating the pain in the 24-week follow-up.

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$p = 0.59$; $I^2 = 9\%$) (Figure S15, Data S2). Both the pulsed ablation technique (SMD = 0.20; 95% CI = −0.19 to 0.59; $p = 0.23$; $I^2 = 30\%$) and CN (SMD = −0.02; 95% CI = −0.34 to 0.30; $p = 0.89$) did not show differences between the groups. Sensitivity analyses did not reveal any relevant variations in the overall results.

3.7.2 | 12-Week Follow-Up

Three studies involving 269 participants were included in this meta-analysis for this outcome [18–20]. The overall analysis did not demonstrate a statistically significant difference between the ablation and sham groups (SMD = 0.13; 95% CI = −0.11 to 0.38; $p = 0.29$; $I^2 = 0\%$) (Figure S16, Data S2). Both the pulsed ablation technique (SMD = 0.13; 95% CI = −0.26 to 0.52; $p = 0.51$; $I^2 = 29\%$) and CN (SMD = 0.14; 95% CI = −0.19 to 0.46; $p = 0.41$) also did not present differences between the groups. Sensitivity analyses did not indicate any relevant variations in the overall result.

3.8 | Adverse Events

The meta-analysis was not feasible due to the absence of quantitative data on adverse events in the included studies. In the studies by Choi et al. [7] and Kwon et al. [21], transient pain or discomfort during cannula insertion or periosteal contact was reported, but the pain was tolerable and did not require additional medication. There were no reports of severe events such as burns, paresthesia, or motor weakness. Radnovich et al. [20] observed various adverse events, such as local pain, bruising, and swelling, with a higher incidence in the intervention group than in the control group. The study by Tabasco et al. [16] described good treatment tolerance, except for an isolated case of resting pain after the procedure. In the studies by Carpenedo et al. [18] and Uematsu et al. [19], no severe complications were reported, although some patients experienced discomfort with needle positioning. Malaithong et al. [17] reported a case of significant swelling, which resolved within 4 weeks and was associated with tendinitis, with no other relevant adverse events.

3.9 | Bias Risk in the Included Studies

The risk of bias in the included studies is summarized in Figure 5.

3.9.1 | Randomization Process

The randomization methodology was evaluated as satisfactory in six studies reviewed [7, 17–21]. However, one study lacked a comprehensive description of its randomization procedure; leading to its classification under the category of presenting some concerns regarding its execution [16].

3.9.2 | Intervention

We assessed a low risk of bias due to deviations in intervention in six studies [7, 17–21]. Nonetheless, one study exhibited signs of deviation from the pre-specified interventions, which resulted in it being categorized as having some concerns regarding these deviations [16].

3.9.3 | Results

Six out of seven studies were deemed satisfactory and thus were considered to present a low risk in this aspect, according to RoB 2 [7, 17–21]. However, one study raised some concerns due to the loss of patient follow-up and the absence of their data, affecting the reliability of its results [16].

3.9.4 | Outcomes

We assessed a low risk of bias in item four of the RoB2 in six studies [7, 16–18, 20, 21]. However, one study identified some concerns in this area due to issues related to the measurement of outcomes [19].

3.9.5 | Selective Reporting

Three articles were classified as having a low risk of selective reporting [7, 18, 21]. Four articles were classified with some concerns due to insufficient information regarding the clinical trial protocol [16, 17, 19, 20].

4 | Discussion

This systematic review focuses exclusively on RCTs that utilize sham interventions as comparators, setting it apart from prior reviews that included a broader range of control types to evaluate the efficacy of ATs [22–24]. By narrowing the focus to sham-controlled studies, this review aims to clarify the uncertainty surrounding the benefits of ablation and identify the populations for whom it may be most effective. The objective is to determine the magnitude of the intervention's effect through simulated procedures, providing a more precise understanding of its efficacy.

The findings demonstrate that ATs effectively reduce pain in OA patients, with CN showing the best results up to 24 weeks despite a general decline in efficacy over time. No significant improvement in QoL was observed, except for isolated cases with CN and the cooled technique. Adverse events were mild and transient, indicating good tolerability.

A previous meta-analysis demonstrated that ATs provide medium-term pain relief, lasting up to 6 months, in patients with chronic knee OA, with no significant long-term effects observed beyond 12 months [22]. These findings are consistent with the results of the present review, which identified significant pain reduction within 6 months. This review analyzed different AT modalities, each characterized by distinct temperatures and mechanisms of action. Despite the improvements reported in most studies, no AT demonstrated clear superiority. Although CN showed long-term pain reduction in one isolated study, its conclusions are not generalizable. In this context, the meta-analysis by Chou et al. [23] and Liu et al. [24] compared ablative methods for knee OA

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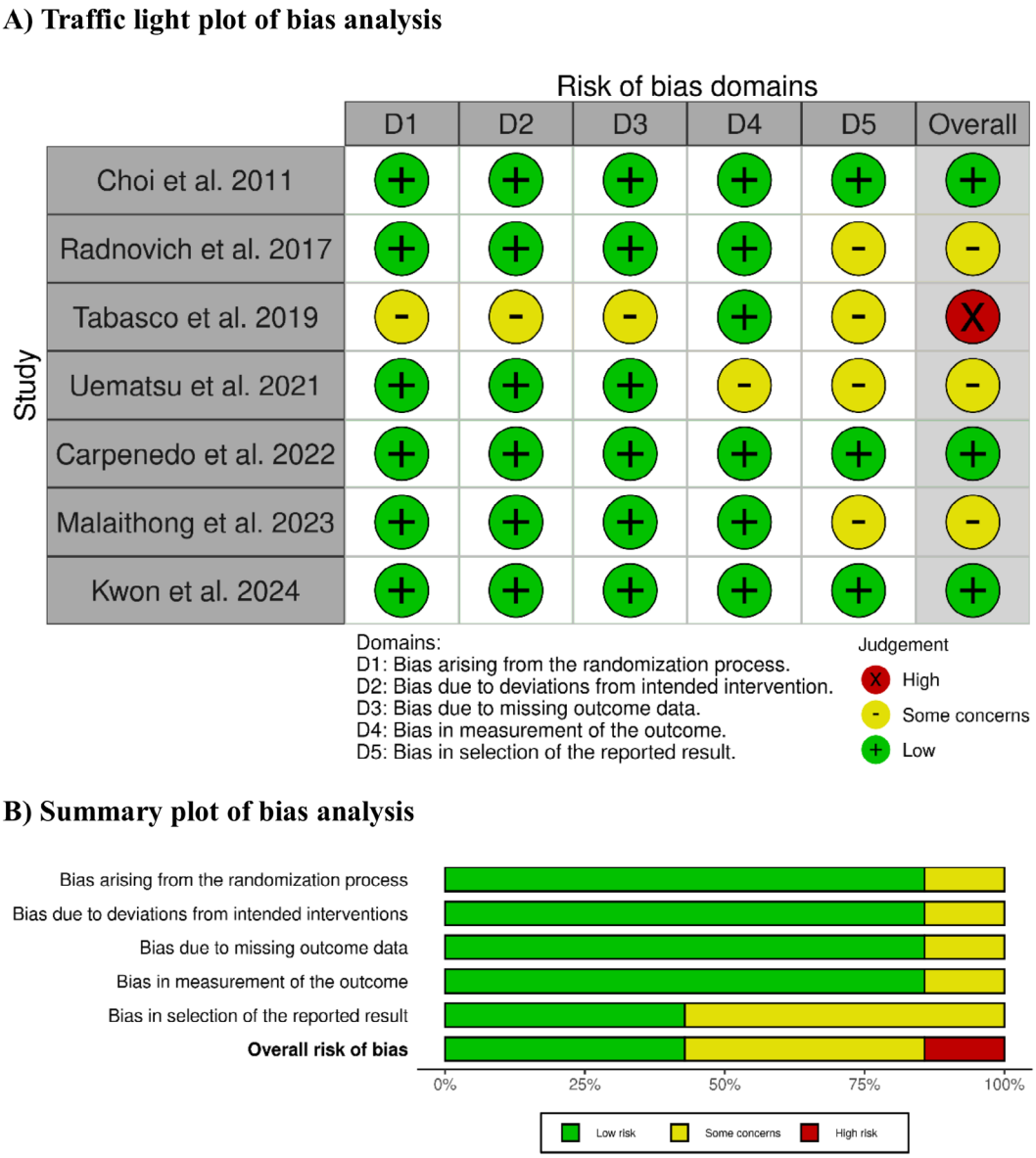

FIGURE 5 | Risk of bias of included studies. (A) Traffic light plot of bias analysis by domain for each included study. (B) Summary plot of bias analysis across all studies.

and concluded that, despite differences in mechanisms of action, no technique demonstrated superior efficacy. Therefore, although ATs demonstrate efficacy compared to placebo for medium-term pain relief, the results among the different methods have shown variations. These findings highlight the need for further studies with larger populations and long-term follow-up to strengthen the available evidence.

The ATs used for pain management in knee OA operate through distinct mechanisms of action, directly influencing their efficacy. Conventional RF employs high temperatures to create permanent lesions on target nerves, disrupting pain transmission [25]. Although effective, this technique can damage adjacent tissues, including motor and sensory fibers, increasing the risk of complications such as neuromas and neuropathic pain [26]. Pulsed RF operates at lower temperatures than continuous RF, delivering energy in intermittent pulses and modulating nerve excitability without causing permanent damage [27]. This approach is less invasive and carries a lower risk of complications, although its analgesic effects are less intense and shorter in duration [28]. Cooled RF combines high temperatures with active electrode cooling, creating more extensive and more precise lesions [29]. CN, in contrast, employs controlled freezing to perform selective neuroablation while preserving structures such as connective tissue and the myelin sheath [30]. This preservation promotes more organized nerve regeneration and reduces complications such as neuromas and neuropathic pain [30]. The extremely low temperatures of CN enable the selective deactivation of nociceptive fibers while preserving motor and sensory fibers not involved in pain transmission [30]. These characteristics may account for the superior results of CN observed in individual studies, demonstrating more effective and prolonged analgesia. In the present study, ATs showed pain relief in patients with knee OA. While CN exhibited promising potential, further research is necessary to validate these findings. Thus, the choice of technique should be guided by a thorough

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clinical evaluation and the expertise of the healthcare professional, highlighting the importance of individualized pain management for OA-related symptoms.

The heterogeneity analysis observed in this review suggests significant differences between the studies, possibly related to the different ATs used, patient population characteristics, and assessment methods employed. The funnel plots showed an asymmetric distribution, possibly indicating the presence of publication bias. These observations may limit the generalization of the results and highlight the need for standardization in intervention methods and inclusion criteria in future research. Therefore, understanding and controlling the factors contributing to this variability will be essential for optimizing pain management guidelines using ATs in OA patients.

When evaluating QoL, the results of this review align with the existing literature. A multicenter randomized clinical trial showed that cooled RF ablation significantly improved QoL compared to standard medical treatment 3 months after the procedure [31]. However, our results did not indicate a significant impact of ATs on QoL, even with pain reduction outcomes. This finding may suggest that, in patients with chronic OA, pain relief does not necessarily translate into noticeable improvements in daily well-being, possibly due to the complexity of functional limitations and the multifactorial nature of the disease. The heterogeneity in the included studies, with variations between techniques and population characteristics, may have influenced these results, underscoring the need for more uniform and standardized trials to assess the impact of these interventions on QoL more accurately.

The results of our review contrast with findings in the literature regarding patient perception. The perception of a procedure is crucial to treatment success as it influences adherence and patient satisfaction [32]. In chronic conditions like OA, this perception goes beyond mere pain reduction, encompassing expectations of improvements in QoL and long-term physical function [33]. A retrospective study by Hong et al. [34], which compared RF thermocoagulation and pulsed RF with steroid injections in treating OA-related pain, showed that after 1 month, there was no significant difference in patient perception. However, at 3 and 6 months, perception was significantly better in the groups receiving RF thermocoagulation. Additionally, a meta-analysis by Liu et al. [24] confirmed that RF significantly improved patient satisfaction levels. Sensitivity analysis, excluding the study by Malaithong et al. [17], resulted in a significant difference, suggesting that heterogeneity among techniques and populations may have influenced the results. The analysis of the studies demonstrated that while ATs showed potential for pain relief, patient perception of the treatment did not significantly follow these improvements.

The adverse events associated with ATs for chronic pain relief, as reported in the literature, are consistent with the findings of this review. In the study by Bergamaschi et al. [35], which compared pulsed RF and suprascapular nerve block, the most common adverse events in the RF group were local pain and bruising. Similarly, Vanneste et al. [36], in a double-blind clinical trial on conventional and cooled RF for chronic knee pain, observed mild adverse events such as temporary pain exacerbation in five patients in the conventional RF group and six cases of subcutaneous bruising and infrapatellar hypoesthesia in the cooled RF group, all of which resolved spontaneously. Overall, the adverse events observed in the studies included in this review were mild, with no severe complications requiring or justifying treatment discontinuation. Reports of transient pain, bruising, and swelling were standard but generally tolerable. No serious events, such as burns or motor weakness, were observed. Therefore, these results suggest that, despite being transient, RF is associated with local adverse events without significant severity.

5 | Study Limitations

This study has some limitations. The small number of included studies and the limited sample size restrict the generalizability of the findings. The heterogeneity in ATs and the high variability in outcomes, such as pain reduction, hinder the comparability of results. The funnel plot indicated potential publication bias; sensitivity analysis revealed that specific studies influence the outcomes. Additionally, variations in follow-up times between studies complicate the assessment of long-term efficacy. To address these limitations, stratified analyses were performed based on technique and follow-up time, with pain and functionality measures standardization. Sensitivity analysis was conducted to verify the robustness of the results by excluding individual studies, and the funnel plot was used to investigate publication bias.

6 | Conclusion

Our study demonstrated that ablation effectively reduces pain, as assessed by VAS, over a follow-up period of up to 24 weeks. Cryoneurolysis showed the most consistent results, while other techniques, such as pulsed and resistive ablation, exhibited variable efficacy. The WOMAC score indicated moderate improvement, with the most significant benefit in patients undergoing cryoneurolysis. However, QoL and patient perception findings were less consistent, with only a few techniques showing significant improvements.

Author Contributions

C.R.A.J. and R.C.O.P. contributed to the conception and design of the study, performed the literature search, and led the drafting of the manuscript. E.S.R.B. and J.P.F.G. participated in study selection and data extraction and contributed to data analysis. L.G.A. assisted with statistical analysis, data synthesis, and drafting of the results section. I.G.K. contributed to the methodological approach and interpretation of findings. L.M.T.A.A., L.E.F.L.-K., and D.C.K. provided critical supervision of the project, contributed to the study design, and critically reviewed the manuscript for important intellectual content. All authors read and approved the final version of the manuscript.

Acknowledgments

The authors have nothing to report.

Ethics Statement

The authors have nothing to report.

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Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data supporting this meta-analysis are derived from previously published studies and are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information. **Figure S1:** Funnel plot of the pain in the 4-week follow-up. **Figure S2:** Forest plot of leave-one-out sensitivity analysis of pain in the 4-week follow-up, excluding Radnovich et al. (2017). **Figure S3:** Funnel plot of the pain in the 12-week follow-up. **Figure S4:** Funnel plot of the pain in the 24-week follow-up. **Figure S5:** Forest plot of leave-one-out sensitivity analysis of pain in the 24-week follow-up, excluding Radnovich et al. (2017). **Figure S6:** Forest plot of WOMAC Score in the 4-week follow-up. **Figure S7:** Forest plot of leave-one-out sensitivity analysis of WOMAC in the 4-week follow-up, excluding Radnovich et al. 2024. **Figure S8:** Forest plot of WOMAC Score in the 12-week follow-up. **Figure S9:** Forest plot of leave-one-out sensitivity analysis of WOMAC in the 12-week follow-up, excluding Radnovich et al. 2017. **Figure S10:**. Forest plot of WOMAC Score in the 24-week follow-up. **Figure S11:** Forest plot of leave-one-out sensitivity analysis of WOMAC in the 24-week follow-up, excluding Radnovich et al. 2017. **Figure S12:** Forest plot of Patient Perception in the 4-week follow-up. **Figure S13:** Forest plot of leave-one-out sensitivity analysis of Patient Perception in the 4-week follow-up, excluding Malaithong et al. 2023. **Figure S14:** Forest plot of Patient Perception in the 12-week follow-up. **Figure S15:** Forest plot of Quality of Life in the 4-week follow-up. **Figure S16:** Forest plot of Quality of Life in the 12-week follow-up.

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